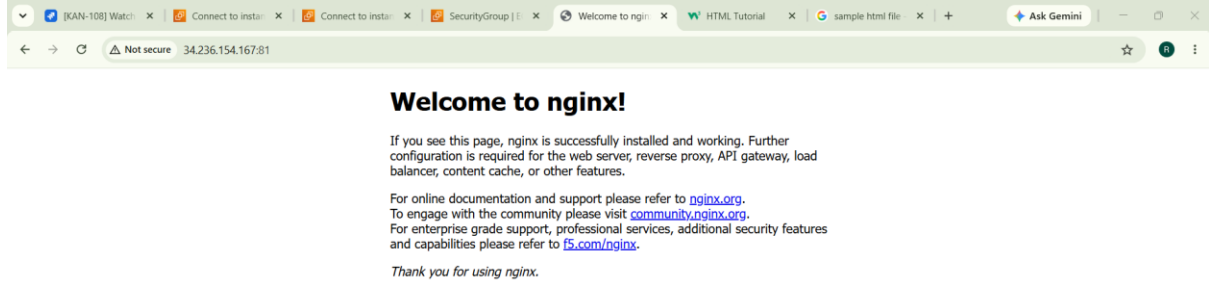


1. Install nginx and run nginx on port number 81.

- ➔ Installed nginx (yum install nginx -y)
- ➔ Started the service (systemctl start nginx)
- ➔ Check status (Active(running))
- ➔ Crosscheck in browser using port number >> publicIP:80
- ➔ Change port number to 81 in (/etc/nginx/nginx.conf) and save the configuration file
- ➔ Add port 81 to the security inbound rules in EC2 instance
- ➔ Restart nginx >> systemctl restart nginx
- ➔ Check status >> (Active (running))
- ➔ Open browser and check with >> Public IP:81

```
[root@ip-172-31-25-75 html]# lsof -i tcp:80
[root@ip-172-31-25-75 html]# lsof -i tcp:81
COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME
nginx 9451 root 8u IPv4 74196 0t0 TCP *:81 (LISTEN)
nginx 9451 root 9u IPv6 74197 0t0 TCP *:81 (LISTEN)
nginx 9452 nginx 8u IPv4 74196 0t0 TCP *:81 (LISTEN)
nginx 9452 nginx 9u IPv6 74197 0t0 TCP *:81 (LISTEN)
nginx 9453 nginx 8u IPv4 74196 0t0 TCP *:81 (LISTEN)
nginx 9453 nginx 9u IPv6 74197 0t0 TCP *:81 (LISTEN)
[root@ip-172-31-25-75 html]#
```



Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

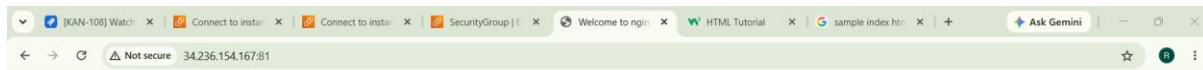


2. Deploy a sample index.html file on nginx.

- ➔ Create a html file: nano /usr/share/nginx/html/index.html
- ➔ Add sample html content and save it.
- ➔ Restart Nginx >> systemctl restart nginx
- ➔ Verify status >> systemctl status nginx
- ➔ Access the page in the browser >> Publicserver IP:81

```
[root@ip-172-31-25-75 html]# pwd
/usr/share/nginx/html
[root@ip-172-31-25-75 html]# cd /usr/share/nginx/html
[root@ip-172-31-25-75 html]# ls
404.html  50x.html  icons  index.html  nginx-logo.png  poweredby.png
[root@ip-172-31-25-75 html]# mv index.html index.html.backup
[root@ip-172-31-25-75 html]# cat index.html.backup
<!DOCTYPE html>
<html>
<head>
<title>welcome to nginx</title>
<style>
html { color:scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>welcome to nginx</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>
<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f3.com/nginx">f3.com/nginx</a>.</p>
<p><em>Thank you for using nginx.</em></p>
</body>
</html>
[root@ip-172-31-25-75 html]# touch index.html
[root@ip-172-31-25-75 html]# vi index.html
[root@ip-172-31-25-75 html]# systemctl restart nginx
[root@ip-172-31-25-75 html]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
   Active: active (running) since wed 2024-08-12 11:53:27 UTC; 39s ago
     Process: 9934 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
     Process: 9935 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
     Process: 9936 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
    Main PID: 9937 (nginx)
       Tasks: 2 (limit: 1059)
        Memory: 3.4M
         CPU: 57ms
    CGroup: /system.slice/nginx.service
            └─9937 "nginx: master process /usr/sbin/nginx"
              └─9938 "nginx: worker process"
                └─9939 "nginx: worker process"

Aug 12 11:53:27 ip-172-31-25-75.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Aug 12 11:53:27 ip-172-31-25-75.ec2.internal nginx[9935]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Aug 12 11:53:27 ip-172-31-25-75.ec2.internal nginx[9935]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Aug 12 11:53:27 ip-172-31-25-75.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-172-31-25-75 html]#
```



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.

Thank you for using nginx.



3. Install Apache and run Apache on port number 82

- ➔ Installed Apache (yum install apache -y)
- ➔ Started the service (systemctl start apache)
- ➔ Check status (Active(running))
- ➔ Crosscheck in browser using port number >> publicIP:80
- ➔ Find path of configuration file (find / -name httpd.conf)
- ➔ Change port number to 82 in .conf and save the configuration file
- ➔ Add port 82 to the security inbound rules in EC2 instance
- ➔ Restart apache >> systemctl restart apache
- ➔ Check status >> (Active (running))
- ➔ Open browser and check with >> Public IP:82

```
pooji@Pooji MINGW64 ~/Downloads
$ scp -i mytest_key.pem ec2-user@98.81.168.208
usage: scp [-346ABCOpqRrStV] [-c cipher] [-D sftp_server_path] [-F ssh_config]
          [-i identity_file] [-J destination] [-l limit]
          [-o ssh_option] [-P port] [-S program] source ... target

pooji@Pooji MINGW64 ~/Downloads
$ ssh -i mytest_key.pem ec2-user@98.81.168.208
The authenticity of host '98.81.168.208 (98.81.168.208)' can't be established.
ED25519 key fingerprint is SHA256:+TW5neGiy8VKjXA170L25H8tP2CuwAnqJHj6XtevNoE.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '98.81.168.208' (ED25519) to the list of known hosts.

#####
#                                     Amazon Linux 2023
#####
#                                     https://aws.amazon.com/linux/amazon-linux-2023
#####

Last login: Wed Aug 12 09:18:30 2026 from 18.206.107.29
[ec2-user@ip-172-31-25-75 ~]$ sudo su -
[root@ip-172-31-25-75 ~]# yum install httpd
Amazon Linux 2023 Kernel Livepatch repository                               467 kB/s | 69 kB    00:00
^[[BDependencies resolved.

=====
Package                                Architecture  Version                                Repository          Size
=====
Installing:
httpd                                   x86_64        2.4.68-1.amzn2023.0.1                 amazonlinux          46 k
Installing dependencies:
apr                                     x86_64        1.7.5-1.amzn2023.0.4                 amazonlinux          129 k
apr-util                               x86_64        1.6.3-1.amzn2023.0.2                 amazonlinux          97 k
apr-util-ldap                          x86_64        1.6.3-1.amzn2023.0.2                 amazonlinux          13 k
generic-logos-httpd                   noarch        18.0.0-12.amzn2023.0.3               amazonlinux          19 k
httpd-core                             x86_64        2.4.68-1.amzn2023.0.1                 amazonlinux          1.4 M
httpd-filesystem                       noarch        2.4.68-1.amzn2023.0.1                 amazonlinux          12 k
httpd-tools                            x86_64        2.4.68-1.amzn2023.0.1                 amazonlinux          80 k
libbrotli                              x86_64        1.0.9-4.amzn2023.0.2                 amazonlinux          315 k
mailcap                                noarch        2.1.49-3.amzn2023.0.3                 amazonlinux          33 k
Installing weak dependencies:
apr-util-openssl                       x86_64        1.6.3-1.amzn2023.0.2                 amazonlinux          15 k
mod_http2                              x86_64        2.0.42-1.amzn2023.0.1                 amazonlinux          167 k
mod_lua                                x86_64        2.4.68-1.amzn2023.0.1                 amazonlinux          59 k

Transaction Summary
=====
Install 13 Packages

Total download size: 2.4 M
Installed size: 7.0 M
Is this ok [y/N]: y
Is this ok [y/N]: y
Downloading Packages:
```

```
(9/13): httpd-tools-2.4.68-1.amzn2023.0.1.x86_64.rpm      2.2 MB/s | 80 kB    00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm      8.2 MB/s | 315 kB   00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm       1.1 MB/s | 33 kB    00:00
(12/13): mod_http2-2.0.42-1.amzn2023.0.1.x86_64.rpm    4.9 MB/s | 167 kB   00:00
(13/13): mod_lua-2.4.68-1.amzn2023.0.1.x86_64.rpm      2.3 MB/s | 59 kB    00:00
-----
Total                                                    11 MB/s | 2.4 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing                                     1/1
  Installing                                    1/13
    : apr-1.7.5-1.amzn2023.0.4.x86_64
  Installing                                    2/13
    : apr-util-1mdb-1.6.3-1.amzn2023.0.2.x86_64
  Installing                                    3/13
    : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  Installing                                    4/13
    : apr-util-1.6.3-1.amzn2023.0.2.x86_64
  Installing                                    5/13
    : mailcap-2.1.49-3.amzn2023.0.3.noarch
  Installing                                    6/13
    : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
  Installing                                    7/13
    : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  Running scriptlet: httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch 8/13
  Installing                                    8/13
    : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
  Installing                                    9/13
    : httpd-core-2.4.68-1.amzn2023.0.1.x86_64
  Installing                                    10/13
    : mod_http2-2.0.42-1.amzn2023.0.1.x86_64
  Installing                                    11/13
    : mod_lua-2.4.68-1.amzn2023.0.1.x86_64
  Installing                                    12/13
    : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  Installing                                    13/13
    : httpd-2.4.68-1.amzn2023.0.1.x86_64
  Running scriptlet: httpd-2.4.68-1.amzn2023.0.1.x86_64 13/13
  Verifying                                    1/13
    : apr-1.7.5-1.amzn2023.0.4.x86_64
  Verifying                                    2/13
    : apr-util-1.6.3-1.amzn2023.0.2.x86_64
  Verifying                                    3/13
    : apr-util-1mdb-1.6.3-1.amzn2023.0.2.x86_64
  Verifying                                    4/13
    : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  Verifying                                    5/13
    : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
  Verifying                                    6/13
    : httpd-2.4.68-1.amzn2023.0.1.x86_64
  Verifying                                    7/13
    : httpd-core-2.4.68-1.amzn2023.0.1.x86_64
  Verifying                                    8/13
    : httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
  Verifying                                    9/13
    : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64
  Verifying                                    10/13
    : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  Verifying                                    11/13
    : mailcap-2.1.49-3.amzn2023.0.3.noarch
  Verifying                                    12/13
    : mod_http2-2.0.42-1.amzn2023.0.1.x86_64
  Verifying                                    13/13
    : mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Installed:
  apr-1.7.5-1.amzn2023.0.4.x86_64          apr-util-1.6.3-1.amzn2023.0.2.x86_64
  apr-util-1mdb-1.6.3-1.amzn2023.0.2.x86_64  apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
  generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch httpd-2.4.68-1.amzn2023.0.1.x86_64
  httpd-core-2.4.68-1.amzn2023.0.1.x86_64  httpd-filesystem-2.4.68-1.amzn2023.0.1.noarch
  httpd-tools-2.4.68-1.amzn2023.0.1.x86_64  libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  mailcap-2.1.49-3.amzn2023.0.3.noarch      mod_http2-2.0.42-1.amzn2023.0.1.x86_64
  mod_lua-2.4.68-1.amzn2023.0.1.x86_64

Complete!
[root@ip-172-31-25-75 ~]# systemctl status httpd
○ httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:httpd.service(8)
[root@ip-172-31-25-75 ~]#
```

```
[root@ip-172-31-25-75 ~]# find / -name httpd.conf
[root@ip-172-31-25-75 ~]# find / -name httpd.conf
/etc/httpd/conf/httpd.conf
/usr/lib/tmpfiles.d/httpd.conf
/usr/lib/sysusers.d/httpd.conf
/opt/httpd.conf
[root@ip-172-31-25-75 ~]# vi /etc/httpd/conf/httpd.conf
[root@ip-172-31-25-75 ~]# systemctl restart httpd
[root@ip-172-31-25-75 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Thu 2026-08-13 10:59:31 UTC; 5s ago
     Docs: man:httpd.service(8)
  Main PID: 2588 (httpd)
    Status: "Started, listening on: port 82"
    Tasks: 177 (limit: 1059)
   Memory: 13.9M
      CPU: 64ms
   CGroup: /system.slice/httpd.service
           └─2588 /usr/sbin/httpd -DFOREGROUND
             └─2589 /usr/sbin/httpd -DFOREGROUND
               └─2590 /usr/sbin/httpd -DFOREGROUND
                 └─2591 /usr/sbin/httpd -DFOREGROUND
                   └─2708 /usr/sbin/httpd -DFOREGROUND

Aug 13 10:59:31 ip-172-31-25-75.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 13 10:59:31 ip-172-31-25-75.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 13 10:59:31 ip-172-31-25-75.ec2.internal httpd[2588]: Server configured, listening on: port 82
[root@ip-172-31-25-75 ~]#
```

Welcome to My Website

This is a sample paragraph explaining the basics of HTML.

Useful Features

- Easy to read structure
- Fully responsive design compatibility
- Clean code template

To learn more, check out the [W3Schools HTML Tutorial](#).

4. Deploy a sample index.html file on Apache.

- ➔ Create a html file: nano /var/www/html/index.html
- ➔ Add sample html content and save it.
- ➔ Restart >> systemctl restart apache
- ➔ Verify status >> systemctl status apache
- ➔ Access the page in the browser >> Public IP:82

```
[root@ip-172-31-25-75 html]# vi index.html
[root@ip-172-31-25-75 html]# cat index.html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Sample Web Page</title>
</head>
<body>

  <h1>Welcome to My Website</h1>
  <p>this is a sample paragraph explaining the basics of HTML.</p>

  <h2>Useful Features</h2>
  <ul>
    <li>Easy to read structure</li>
    <li>Fully responsive design compatibility</li>
    <li>Clean code template</li>
  </ul>

  <p>To learn more, check out the <a href="https://www.w3schools.com/html/" target="_blank">W3Schools HTML Tutorial</a>.</p>
</body>
</html>
[root@ip-172-31-25-75 html]# |
```



5. Install Apache tomcat on port number 8082.

- ➔ Followed the process from the video
- ➔ Install Java: yum install java.
- ➔ Download and extract Apache Tomcat : `wget <url- of-tomcat-tar-file>.`
- ➔ Install java if not available
- ➔ Find the Java installation path `>> readlink -f $(which java)`
- ➔ Set JAVA_HOME `>> export JAVA_HOME=/usr/lib/jvm/java-17-amazon-corretto`
- ➔ Start using startup file `>> bash startup.sh`
- ➔ Check status by `>> ps -ef | grep tomcat`
- ➔ Check in browser with Public IP:8080
- ➔ Edit Tomcat configuration: `vi /opt/tomcat/conf/server.xml`
- ➔ Change the connector port from 8080 to 8082: `<Connector port="8082" ... />`
- ➔ Start Tomcat: `/opt/tomcat/bin/startup.sh`
- ➔ Verify `>> lsof -i tcp:8082`
- ➔ Check in the browser: Public IP:8082

```
Verifying
Installing
catro-1.18.0-4.amzn2023.0.3.x86_64
dejava-sans-fonts-2.37-16.amzn2023.0.2.noarch
fontconfig-2.13.94-2.amzn2023.0.2.x86_64
glib-5.2.1-9.amzn2023.0.4.x86_64
glib-5.2.1-9.amzn2023.0.4.x86_64
graphite2-1.1.14-7.amzn2023.0.3.x86_64
java-17-amazon-corretto-headless-1:17.0.20-8-1.amzn2023.1.x86_64
libICE-1.1.1-3.amzn2023.0.1.x86_64
libX11-common-1.8.10-2.amzn2023.0.1.noarch
libX11-1.8.2-1.amzn2023.0.1.x86_64
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64
libXau-1.0.11-6.amzn2023.0.1.x86_64
libXdmcp-1.1.5-6.amzn2023.0.1.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libpng-2:1.6.37-10.amzn2023.0.13.x86_64
xul-common-0.4.3-16.amzn2023.0.2.noarch
catro-1.18.0-4.amzn2023.0.3.x86_64
dejava-sans-fonts-2.37-16.amzn2023.0.2.noarch
fontconfig-2.13.94-2.amzn2023.0.2.x86_64
glib-5.2.1-9.amzn2023.0.4.x86_64
glib-5.2.1-9.amzn2023.0.4.x86_64
graphite2-1.1.14-7.amzn2023.0.3.x86_64
java-17-amazon-corretto-headless-1:17.0.20-8-1.amzn2023.1.x86_64
libICE-1.1.1-3.amzn2023.0.1.x86_64
libX11-common-1.8.10-2.amzn2023.0.1.noarch
libX11-1.8.2-1.amzn2023.0.1.x86_64
libXext-1.3.6-1.amzn2023.0.1.x86_64
libXrender-0.9.11-6.amzn2023.0.1.x86_64
libjpeg-turbo-2.1.4-2.amzn2023.0.5.x86_64
pixman-0.43.4-1.amzn2023.0.4.x86_64
libXau-1.0.11-6.amzn2023.0.1.x86_64
libXdmcp-1.1.5-6.amzn2023.0.1.x86_64
libXt-1.3.0-3.amzn2023.0.1.x86_64
libpng-2:1.6.37-10.amzn2023.0.13.x86_64
xul-common-0.4.3-16.amzn2023.0.2.noarch
Complete!
[root@ip-172-31-25-75 bin]# java -version
openjdk version "17.0.20" 2025-07-21 LTS
OpenJDK Runtime Environment Corretto-17.0.20.8.1 (build 17.0.20.8.1)
OpenJDK 64-Bit Server VM Corretto-17.0.20.8.1 (build 17.0.20.8.1, mixed mode, sharing)
[root@ip-172-31-25-75 bin]# readlink -f $(which java)
/usr/lib/jvm/java-17-amazon-corretto-x86_64/bin/java
[root@ip-172-31-25-75 bin]# export JAVA_HOME=/usr/lib/jvm/java-17-amazon-corretto
[root@ip-172-31-25-75 bin]# echo $JAVA_HOME
/usr/lib/jvm/java-17-amazon-corretto
[root@ip-172-31-25-75 bin]# bash startup.sh
bash: startup.sh: No such file or directory
[root@ip-172-31-25-75 bin]# bash startup.sh
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.120
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.120
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.120/temp
Using JRE_HOME:         /usr/lib/jvm/java-17-amazon-corretto
Using CLASSPATH:        /opt/apache-tomcat-9.0.120/bin/bootstrap.jar:/opt/apache-tomcat-9.0.120/bin/tomcat-juli.jar
Using CATALINA_OPTS:
tomcat started.
[root@ip-172-31-25-75 bin]# ps -ef | grep tomcat
root      1951      1  0:23 pts/1    00:00:04 /usr/lib/jvm/java-17-amazon-corretto/bin/java -Djava.util.logging.config.file=/opt/apache-tomcat-9.0.120/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Djdk.tls.cipherSuites=3048 -Djava.protocol.handler.pkgs=org.apache.catalina.webresources -Dsun.io.useCanonCaches=false -Dorg.apache.catalina.security.SecurityListener=UMASK-0027 -Dignore.endorsed.dirs=-classpath /opt/apache-tomcat-9.0.120/bin/bootstrap.jar:/opt/apache-tomcat-9.0.120/bin/tomcat-juli.jar -Dcatalina.base=/opt/apache-tomcat-9.0.120 -Dcatalina.home=/opt/apache-tomcat-9.0.120 -Djava.io.tmpdir=/opt/apache-tomcat-9.0.120/temp org.apache.catalina.startup.Bootstrap start
root      3043    2088  0 07:23 pts/1    00:00:00 grep --color=auto tomcat
[root@ip-172-31-25-75 bin]# netstat -na | grep 8080
tcp        0      0 0.0.0.0:8080          *LISTEN*
[root@ip-172-31-25-75 bin]# ls
bootstrap.jar  catalina.sh  commons-daemon-native.tar.gz  configtest.sh  digest.sh  setclasspath.bat  shutdown.sh  tomcat-juli.jar  tool-wrapper.sh
catalina-tasks.xml  ciphers.bat  commons-daemon.jar  daemon.sh  makebase.bat  setclasspath.sh  startup.bat  tomcat-native.tar.gz  version.bat
catalina.bat  configtest.bat  digest.bat  makebase.sh  shutdown.bat  startup.sh  tool-wrapper.bat  version.sh
-bash: cs: command not found
[root@ip-172-31-25-75 bin]# ls
bootstrap.jar  catalina.sh  commons-daemon-native.tar.gz  configtest.sh  digest.sh  setclasspath.bat  shutdown.sh  tomcat-juli.jar  tool-wrapper.sh
catalina-tasks.xml  ciphers.bat  commons-daemon.jar  daemon.sh  makebase.bat  setclasspath.sh  startup.bat  tomcat-native.tar.gz  version.bat
catalina.bat  configtest.bat  digest.bat  makebase.sh  shutdown.bat  startup.sh  tool-wrapper.bat  version.sh
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  temp  webapps  work
[root@ip-172-31-25-75 bin]#
```

← → ↻ Not secure 54.226.96.44:8082

Home Documentation Configuration Examples Wiki Mailing Lists Find Help

Apache Tomcat/9.0.120

If you're seeing this, you've successfully installed Tomcat. Congratulations!



Recommended Reading:

- [Security Considerations How-To](#)
- [Manager Application How-To](#)
- [Clustering/Session Replication How-To](#)

Server Status
Manager App
Host Manager

Developer Quick Start

Tomcat Setup First Web Application	Realms & AAA JDBC Data Sources	Examples	Servlet Specifications Tomcat Versions
---------------------------------------	-----------------------------------	----------	---

Managing Tomcat

For security, access to the `manager.webapp` is restricted. Users are defined in:

```
$CATALINA_HOME/conf/tomcat-users.xml
```

In Tomcat 9.0 access to the manager application is split between different users.
[Read more...](#)

[Release Notes](#)
[Changelog](#)
[Migration Guide](#)
[Security Notices](#)

Documentation

[Tomcat 9.0 Documentation](#)
[Tomcat 9.0 Configuration](#)
[Tomcat Wiki](#)

Find additional important configuration information in:

`$CATALINA_HOME/RUNNING.txt`

Developers may be interested in:

- [Tomcat 9.0 Bug Database](#)
- [Tomcat 9.0 JavaDocs](#)
- [Tomcat 9.0 Git Repository at GitHub](#)

Getting Help

FAQ and Mailing Lists

The following mailing lists are available:

- [tomcat-announce](#)
Important announcements, releases, security vulnerability notifications. (Low volume).
- [tomcat-users](#)
User support and discussion
- [taglibs-user](#)
User support and discussion for [Apache Taglibs](#)
- [tomcat-dev](#)
Development mailing list, including commit messages

6. Deploy a sample app on webapps

- ➔ Download a sample app from google to the Webapps Directory
- ➔ Restart tomcat
- ➔ Check in browser using >> Public IP:8082/sample(application name)

← → ↻ Not secure 54.226.96.44:8082/sample/



Sample "Hello, World" Application

This is the home page for a sample application used to illustrate the source directory organization of a web application utilizing the principles outlined in the Application Developer's Guide.

To prove that they work, you can execute either of the following links:

- To a [JSP page](#).
- To a [servlet](#).

7. Create a tomcat.service file for tomcat.

- ➔ Create a tomcat.service file in the service directory (/usr/lib/systemd/system/)
- ➔ Configure Tomcat paths and start/stop commands
- ➔ Relod systemd >> systemctl daemon-reload
- ➔ Enable tomcat >> systemctl enable tomcat
- ➔ Start tomcat >> systemctl start tomcat
- ➔ Check status >> systemctl status tomcat

```

[root@ip-172-31-25-75 ~]# systemctl start tomcat
Starting Tomcat Web Application Container:
Loaded: loaded (/usr/lib/systemd/system/tomcat.service; enabled; preset: disabled)
Active: active (running) since Thu 2026-08-13 07:57:17 UTC; 19s ago
Main PID: 4800 (java)
Tasks: 3
CGroup: /system.slice/tomcat.service
└─ 4800 /usr/bin/java -Djava.util.logging.config.file=/opt/apache-tomcat-9.0.120/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderManager -Djdk.tls.ephemeralDHkeys=
Aug 13 07:57:18 ip-172-31-25-75.ec2.internal systemd[1]: Stopped tomcat.service - Apache Tomcat Web Application Container...
Aug 13 07:57:18 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:18.940 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [/opt/apache-tomcat-9.0.120/webapps/ROOT]
Aug 13 07:57:18 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:18.989 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [/opt/apache-tomcat-9.0.120/webapps/ROOT]
Aug 13 07:57:18 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:19.021 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [/opt/apache-tomcat-9.0.120/webapps/ROOT]
Aug 13 07:57:19 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:19.283 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [/opt/apache-tomcat-9.0.120/webapps/ROOT]
Aug 13 07:57:19 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:19.324 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [/opt/apache-tomcat-9.0.120/webapps/ROOT]
Aug 13 07:57:19 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:19.349 INFO [main] org.apache.catalina.startup.HostConfig.deployDirectory Deploying web application directory [/opt/apache-tomcat-9.0.120/webapps/ROOT]
Aug 13 07:57:19 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:19.354 INFO [main] org.apache.coyote.AbstractProtocol.start Starting ProtocolHandler ["http-nio-8082"]
Aug 13 07:57:19 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:19.370 INFO [main] org.apache.catalina.startup.Catalina.start Server startup in 3703 milliseconds
[root@ip-172-31-25-75 ~]# systemctl stop tomcat
Stopping Tomcat Web Application Container:
Loaded: loaded (/usr/lib/systemd/system/tomcat.service; enabled; preset: disabled)
Active: failed (Result: exit-code) since Thu 2026-08-13 07:57:44 UTC; 3s ago
Duration: 26.996s
Process: 4800 ExecStart=/opt/apache-tomcat-9.0.120/bin/catalina.sh run (code=exited, status=143)
Process: 4844 ExecStop=/opt/apache-tomcat-9.0.120/bin/shutdown.sh (code=exited, status=0/SUCCESS)
Main PID: 4800 (code=exited, status=143)
CGroup: /system.slice/tomcat.service
└─ 4800
Aug 13 07:57:43 ip-172-31-25-75.ec2.internal systemd[1]: Stopped tomcat.service - Apache Tomcat Web Application Container...
Aug 13 07:57:44 ip-172-31-25-75.ec2.internal catallina.sh[4800]: NOTE: Picked up TOMCAT_COMPAT_FORK= --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNAMED
Aug 13 07:57:44 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:44.256 INFO [main] org.apache.catalina.core.StandardService.await A valid shutdown command was received via the shutdown port.
Aug 13 07:57:44 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:44.259 INFO [main] org.apache.coyote.AbstractProtocol.pause Pausing ProtocolHandler ["http-nio-8082"]
Aug 13 07:57:44 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:44.268 INFO [main] org.apache.catalina.core.StandardService.stopInternal Stopping service [Catalina]
Aug 13 07:57:44 ip-172-31-25-75.ec2.internal catallina.sh[4800]: 13-Aug-2026 07:57:44.296 INFO [main] org.apache.coyote.AbstractProtocol.stop Stopping ProtocolHandler ["http-nio-8082"]
Aug 13 07:57:44 ip-172-31-25-75.ec2.internal systemd[1]: tomcat.service: Main process exited, code=exited, status=143/n/a
Aug 13 07:57:44 ip-172-31-25-75.ec2.internal systemd[1]: tomcat.service: Failed with result 'exit-code'.
Aug 13 07:57:44 ip-172-31-25-75.ec2.internal systemd[1]: Stopped tomcat.service - Apache Tomcat Web Application Container.
Aug 13 07:57:44 ip-172-31-25-75.ec2.internal systemd[1]: tomcat.service: Consumed 5.118s CPU time.
[root@ip-172-31-25-75 ~]#

```

8. Configure HA Proxy server

- ➔ Followed the shared document and configured HA Proxy server.

Run following command to Access the Server-1

- `ssh -i your-key.pem ec2-user@Pubipaddress of Server-1 instance`
- `sudo -i`
- `yum update -y`

[illegible]

→ yum install httpd -y

```
[root@ip-172-31-28-41 ~]# yum install httpd -y
Last metadata expiration check: 0:00:58 ago on Thu Aug 13 09:05:44 2026.
Dependencies resolved.
Package Architecture Version Repository Size
Installing:
httpd x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 46 k
Installing dependencies:
apr x86_64 1.7.5-1.amzn2023.0.4 amazonlinux 129 k
apr-util x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 97 k
apr-util-ldap x86_64 1.6.3-1.amzn2023.0.2 amazonlinux 13 k
generic-logos-httpd noarch 18.0.0-12.amzn2023.0.3 amazonlinux 1.4 M
httpd-core x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 80 k
httpdfilesystem x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 12 k
httpd-tools x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 315 k
libbrotli x86_64 1.0.9-4.amzn2023.0.2 amazonlinux 33 k
mailcap noarch 2.1.49-3.amzn2023.0.3 amazonlinux 15 k
mod_http2 x86_64 2.0.42-1.amzn2023.0.2 amazonlinux 167 k
mod_lua x86_64 2.4.68-1.amzn2023.0.1 amazonlinux 59 k
Transaction Summary
Total download size: 2.4 M
Installed size: 7.0 M
Downloading Packages:
(1/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm 333 kB/s | 13 kB 00:00
(2/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm 2.1 MB/s | 97 kB 00:00
(3/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm 474 kB/s | 15 kB 00:00
(4/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 601 kB/s | 1.4 MB 00:00
(5/13): httpd-2.4.68-1.amzn2023.0.1.x86_64.rpm 1.5 MB/s | 46 kB 00:00
(6/13): httpd-core-2.4.68-1.amzn2023.0.1.x86_64.rpm 35 MB/s | 1.4 MB 00:00
(7/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm 1.0 MB/s | 129 kB 00:00
(8/13): httpdfilesystem-2.4.68-1.amzn2023.0.1.noarch.rpm 339 kB/s | 12 kB 00:00
(9/13): httpd-tools-2.4.68-1.amzn2023.0.1.x86_64.rpm 2.8 MB/s | 80 kB 00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm 8.8 MB/s | 315 kB 00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm 1.3 MB/s | 33 kB 00:00
(12/13): mod_http2-2.0.42-1.amzn2023.0.1.x86_64.rpm 5.6 MB/s | 167 kB 00:00
(13/13): mod_lua-2.4.68-1.amzn2023.0.1.x86_64.rpm 2.0 MB/s | 59 kB 00:00
Total 10 MB/s | 2.4 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Installing : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64 2/13
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
Installing : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
Installing : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13
Installing : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64 6/13
(1/13): apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64.rpm 333 kB/s | 13 kB 00:00
(2/13): apr-util-1.6.3-1.amzn2023.0.2.x86_64.rpm 2.1 MB/s | 97 kB 00:00
(3/13): apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64.rpm 474 kB/s | 15 kB 00:00
(4/13): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 601 kB/s | 1.4 MB 00:00
(5/13): httpd-2.4.68-1.amzn2023.0.1.x86_64.rpm 1.5 MB/s | 46 kB 00:00
(6/13): httpd-core-2.4.68-1.amzn2023.0.1.x86_64.rpm 35 MB/s | 1.4 MB 00:00
(7/13): apr-1.7.5-1.amzn2023.0.4.x86_64.rpm 1.0 MB/s | 129 kB 00:00
(8/13): httpdfilesystem-2.4.68-1.amzn2023.0.1.noarch.rpm 339 kB/s | 12 kB 00:00
(9/13): httpd-tools-2.4.68-1.amzn2023.0.1.x86_64.rpm 2.8 MB/s | 80 kB 00:00
(10/13): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm 8.8 MB/s | 315 kB 00:00
(11/13): mailcap-2.1.49-3.amzn2023.0.3.noarch.rpm 1.3 MB/s | 33 kB 00:00
(12/13): mod_http2-2.0.42-1.amzn2023.0.1.x86_64.rpm 5.6 MB/s | 167 kB 00:00
(13/13): mod_lua-2.4.68-1.amzn2023.0.1.x86_64.rpm 2.0 MB/s | 59 kB 00:00
Total 10 MB/s | 2.4 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : 1/1
Installing : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Installing : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64 2/13
Installing : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 3/13
Installing : apr-util-1.6.3-1.amzn2023.0.2.x86_64 4/13
Installing : mailcap-2.1.49-3.amzn2023.0.3.noarch 5/13
Installing : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64 6/13
Installing : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 7/13
Running scriptlet: httpdfilesystem-2.4.68-1.amzn2023.0.1.noarch 8/13
Installing : httpdfilesystem-2.4.68-1.amzn2023.0.1.noarch 8/13
Installing : httpd-core-2.4.68-1.amzn2023.0.1.x86_64 9/13
Installing : mod_http2-2.0.42-1.amzn2023.0.1.x86_64 10/13
Installing : mod_lua-2.4.68-1.amzn2023.0.1.x86_64 11/13
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 12/13
Installing : httpd-2.4.68-1.amzn2023.0.1.x86_64 13/13
Running scriptlet: httpd-2.4.68-1.amzn2023.0.1.x86_64 13/13
Verifying : apr-1.7.5-1.amzn2023.0.4.x86_64 1/13
Verifying : apr-util-1.6.3-1.amzn2023.0.2.x86_64 2/13
Verifying : apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64 3/13
Verifying : apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64 4/13
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 5/13
Verifying : httpd-2.4.68-1.amzn2023.0.1.x86_64 6/13
Verifying : httpd-core-2.4.68-1.amzn2023.0.1.x86_64 7/13
Verifying : httpdfilesystem-2.4.68-1.amzn2023.0.1.noarch 8/13
Verifying : httpd-tools-2.4.68-1.amzn2023.0.1.x86_64 9/13
Verifying : libbrotli-1.0.9-4.amzn2023.0.2.x86_64 10/13
Verifying : mailcap-2.1.49-3.amzn2023.0.3.noarch 11/13
Verifying : mod_http2-2.0.42-1.amzn2023.0.1.x86_64 12/13
Verifying : mod_lua-2.4.68-1.amzn2023.0.1.x86_64 13/13
Installed:
apr-1.7.5-1.amzn2023.0.4.x86_64 apr-util-1.6.3-1.amzn2023.0.2.x86_64 apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64 apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch httpd-2.4.68-1.amzn2023.0.1.x86_64 httpd-core-2.4.68-1.amzn2023.0.1.x86_64 httpdfilesystem-2.4.68-1.amzn2023.0.1.noarch
httpd-tools-2.4.68-1.amzn2023.0.1.x86_64 libbrotli-1.0.9-4.amzn2023.0.2.x86_64 mailcap-2.1.49-3.amzn2023.0.3.noarch mod_http2-2.0.42-1.amzn2023.0.1.x86_64
mod_lua-2.4.68-1.amzn2023.0.1.x86_64
Complete!
[root@ip-172-31-28-41 ~]#
```

- vi /etc/hosts
- add HA proxy server Public IP

```
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1        localhost6 localhost6.localdomain6
54.159.48.208 load_balancer
```

```
~
~
~
~
~
~
~
~
```

Complete:

```
[root@ip-172-31-28-41 ~]# vi /etc/hosts
[root@ip-172-31-28-41 ~]# vi /etc/hosts
[root@ip-172-31-28-41 ~]# cat /etc/host
cat: /etc/host: No such file or directory
[root@ip-172-31-28-41 ~]# cat /etc/hosts
127.0.0.1    localhost localhost.localdomain localhost4 localhost4.localdomain4
::1        localhost6 localhost6.localdomain6
54.159.48.208 load_balancer
[root@ip-172-31-28-41 ~]#
```

Run below command on Server-1

ping load_balancer -c 4

```
54.159.48.208 load_balancer
[root@ip-172-31-28-41 ~]# ping load_balancer -c 4
PING load_balancer (54.159.48.208) 56(84) bytes of data.

--- load_balancer ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3104ms

[root@ip-172-31-28-41 ~]#
```

- Systemctl start httpd
- Systemctl status httpd

```
[root@ip-172-31-28-41 ~]# systemctl start httpd
[root@ip-172-31-28-41 ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Thu 2026-08-13 09:11:52 UTC; 6s ago
     Docs: man:httpd.service(8)
  Main PID: 26290 (httpd)
    Status: "Started, listening on: port 80"
    Tasks: 177 (limit: 1059)
   Memory: 13.8M
      CPU: 65ms
   CGroup: /system.slice/httpd.service
           └─26290 /usr/sbin/httpd -DFOREGROUND
             └─26291 /usr/sbin/httpd -DFOREGROUND
               └─26292 /usr/sbin/httpd -DFOREGROUND
                 └─26298 /usr/sbin/httpd -DFOREGROUND
                   └─26314 /usr/sbin/httpd -DFOREGROUND

Aug 13 09:11:52 ip-172-31-28-41.ec2.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Aug 13 09:11:52 ip-172-31-28-41.ec2.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Aug 13 09:11:52 ip-172-31-28-41.ec2.internal httpd[26290]: Server configured, listening on: port 80
[root@ip-172-31-28-41 ~]#
```

← → 🔍 Not secure 54.243.17.85 ☆ ⓘ ⋮

It works!

Run following command to Access Server-2

- ```
root@ip-172-31-28-238 ~# ssh -i mytest_key.pem ec2-user@98.84.180.46
The authenticity of host '98.84.180.46 (98.84.180.46)' can't be established.
ECDSA key fingerprint is SHA256:64shUqZabwWtSwB0Uwvd7Wz4cy3eFwIdwMkKagc.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '98.84.180.46' (ED25519) to the list of known hosts.
ec2-user@ip-172-31-28-238 ~$ sudo -i
root@ip-172-31-28-238 ~# if yum update -y
Amazon Linux 2023 kernel livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
root@ip-172-31-28-238 ~# if yum install nginx -y
Last metadata expiration check: 0:00:19 ago on Thu Aug 13 09:20:54 2026.
Dependencies resolved.

Package Architecture Version Repository Size

Installing:
nginx x86_64 1:1.30.4-1.amzn2023.0.1 amazonlinux 34 k
Installing dependencies:
generic-logos-httpd noarch 18.0-0-12.amzn2023.0.3 amazonlinux 19 k
gperftools-libs-2.9.1-1.amzn2023.0.3 x86_64 2.9.1-1.amzn2023.0.3 amazonlinux 308 k
libunwind-1.4.0-5.amzn2023.0.3 x86_64 1.4.0-5.amzn2023.0.3 amazonlinux 66 k
nginx-core-1.30.4-1.amzn2023.0.1 x86_64 1:1.30.4-1.amzn2023.0.1 amazonlinux 711 k
nginx-filesystem-1.30.4-1.amzn2023.0.1 noarch 1.1.30.4-1.amzn2023.0.1 amazonlinux 10 k
nginx-mimetypes-2.1.49-3.amzn2023.0.3 noarch 2.1.49-3.amzn2023.0.3 amazonlinux 21 k

Transaction Summary

Install 7 Packages
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.9 MB/s | 308 kb 00:00
(2/7): generic-logos-httpd-18.0-0-12.amzn2023.0.3.noarch.rpm 441 kb/s | 19 kb 00:00
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.4 MB/s | 66 kb 00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kb 00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm 418 kb/s | 10 kb 00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm 19 MB/s | 711 kb 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 684 kb/s | 21 kb 00:00

Total
Running transaction check
Total download size: 1.1 M
Installed size: 3.8 M
Transaction Summary

Install 7 Packages
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.9 MB/s | 308 kb 00:00
(2/7): generic-logos-httpd-18.0-0-12.amzn2023.0.3.noarch.rpm 441 kb/s | 19 kb 00:00
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.4 MB/s | 66 kb 00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kb 00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm 418 kb/s | 10 kb 00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm 19 MB/s | 711 kb 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 684 kb/s | 21 kb 00:00

Total
Running transaction check
Total download size: 1.1 M
Installed size: 3.8 M
Transaction Summary

Install 7 Packages
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.9 MB/s | 308 kb 00:00
(2/7): generic-logos-httpd-18.0-0-12.amzn2023.0.3.noarch.rpm 441 kb/s | 19 kb 00:00
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.4 MB/s | 66 kb 00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kb 00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm 418 kb/s | 10 kb 00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm 19 MB/s | 711 kb 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 684 kb/s | 21 kb 00:00

Total
Running transaction check
Total download size: 1.1 M
Installed size: 3.8 M
Transaction Summary

Install 7 Packages
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.9 MB/s | 308 kb 00:00
(2/7): generic-logos-httpd-18.0-0-12.amzn2023.0.3.noarch.rpm 441 kb/s | 19 kb 00:00
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.4 MB/s | 66 kb 00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kb 00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm 418 kb/s | 10 kb 00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm 19 MB/s | 711 kb 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 684 kb/s | 21 kb 00:00

Total
Running transaction check
Total download size: 1.1 M
Installed size: 3.8 M
Transaction Summary

Install 7 Packages
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.9 MB/s | 308 kb 00:00
(2/7): generic-logos-httpd-18.0-0-12.amzn2023.0.3.noarch.rpm 441 kb/s | 19 kb 00:00
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.4 MB/s | 66 kb 00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kb 00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm 418 kb/s | 10 kb 00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm 19 MB/s | 711 kb 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 684 kb/s | 21 kb 00:00

Total
Running transaction check
Total download size: 1.1 M
Installed size: 3.8 M
Transaction Summary

Install 7 Packages
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.9 MB/s | 308 kb 00:00
(2/7): generic-logos-httpd-18.0-0-12.amzn2023.0.3.noarch.rpm 441 kb/s | 19 kb 00:00
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.4 MB/s | 66 kb 00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kb 00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm 418 kb/s | 10 kb 00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm 19 MB/s | 711 kb 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 684 kb/s | 21 kb 00:00

Total
Running transaction check
Total download size: 1.1 M
Installed size: 3.8 M
Transaction Summary

Install 7 Packages
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.9 MB/s | 308 kb 00:00
(2/7): generic-logos-httpd-18.0-0-12.amzn2023.0.3.noarch.rpm 441 kb/s | 19 kb 00:00
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.4 MB/s | 66 kb 00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kb 00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm 418 kb/s | 10 kb 00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm 19 MB/s | 711 kb 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 684 kb/s | 21 kb 00:00

Total
Running transaction check
Total download size: 1.1 M
Installed size: 3.8 M
Transaction Summary

Install 7 Packages
Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.9 MB/s | 308 kb 00:00
(2/7): generic-logos-httpd-18.0-0-12.amzn2023.0.3.noarch.rpm 441 kb/s | 19 kb 00:00
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.4 MB/s | 66 kb 00:00
(4/7): nginx-1.30.4-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kb 00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm 418 kb/s | 10 kb 00:00
(6/7): nginx-core-1.30.4-1.amzn2023.0.1.x86_64.rpm 19 MB/s | 711 kb 00:00
(7/7): nginx-mimetypes-2.1.
```

```
Transaction Summary
Install 7 Packages

Total download size: 1.1 M
Installed size: 3.8 M
Downloading Packages:
(1/7): gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64.rpm 7.9 MB/s | 308 kB 00:00
(2/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch.rpm 441 kB/s | 19 kB 00:00
(3/7): libunwind-1.4.0-5.amzn2023.0.3.x86_64.rpm 1.4 MB/s | 66 kB 00:00
(4/7): nghttp-1.30.4-1.amzn2023.0.1.x86_64.rpm 1.1 MB/s | 34 kB 00:00
(5/7): nginx-filesystem-1.30.4-1.amzn2023.0.1.noarch.rpm 416 kB/s | 10 kB 00:00
(6/7): ngxin-core-1.30.4-1.amzn2023.0.1.x86_64.rpm 19 MB/s | 711 kB 00:00
(7/7): nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch.rpm 684 kB/s | 21 kB 00:00

Total 8.0 MB/s | 1.1 MB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing... :
Running scriptlet: nginx-filesystem-1:1.30.4-1.amzn2023.0.1.noarch 1/1
Installing : nginx-filesystem-1:1.30.4-1.amzn2023.0.1.noarch 1/7
Installing : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 2/7
Installing : libunwind-1.4.0-5.amzn2023.0.3.x86_64 3/7
Installing : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 4/7
Installing : nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64 5/7
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 6/7
Installing : nginx-1:1.30.4-1.amzn2023.0.1.x86_64 7/7
Running scriptlet: nginx-1:1.30.4-1.amzn2023.0.1.x86_64 7/7
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 1/7
Verifying : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 2/7
Verifying : libunwind-1.4.0-5.amzn2023.0.3.x86_64 3/7
Verifying : nginx-1:1.30.4-1.amzn2023.0.1.x86_64 4/7
Verifying : nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64 5/7
Verifying : nginx-filesystem-1:1.30.4-1.amzn2023.0.1.noarch 6/7
Verifying : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 7/7

Installed:
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 libunwind-1.4.0-5.amzn2023.0.3.x86_64 nghttp-1x-1.30.4-1.amzn2023.0.1.x86_64
nginx-core-1:1.30.4-1.amzn2023.0.1.x86_64 nginx-filesystem-1:1.30.4-1.amzn2023.0.1.noarch nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
[root@ip-172-31-28-238 ~]#
```

- systemctl start nginx
- systemctl status nginx

```
[root@ip-172-31-28-238 ~]# systemctl start nginx
[root@ip-172-31-28-238 ~]# systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
 Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: disabled)
 Active: active (running) since Thu 2026-08-13 09:27:48 UTC; 7s ago
 Process: 26713 ExecStartPre=/usr/bin/rm -f /run/nginx.pid (code=exited, status=0/SUCCESS)
 Process: 26714 ExecStartPre=/usr/sbin/nginx -t (code=exited, status=0/SUCCESS)
 Process: 26715 ExecStart=/usr/sbin/nginx (code=exited, status=0/SUCCESS)
 Main PID: 26716 (nginx)
 Tasks: 3 (limit: 1059)
 Memory: 3.7M
 CPU: 57ms
 CGroup: /system.slice/nginx.service
 └─26716 "nginx: master process /usr/sbin/nginx"
 └─26717 "nginx: worker process"
 └─26718 "nginx: worker process"

Aug 13 09:27:48 ip-172-31-28-238.ec2.internal systemd[1]: Starting nginx.service - The nginx HTTP and reverse proxy server...
Aug 13 09:27:48 ip-172-31-28-238.ec2.internal nginx[26714]: nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
Aug 13 09:27:48 ip-172-31-28-238.ec2.internal nginx[26714]: nginx: configuration file /etc/nginx/nginx.conf test is successful
Aug 13 09:27:48 ip-172-31-28-238.ec2.internal systemd[1]: Started nginx.service - The nginx HTTP and reverse proxy server.
[root@ip-172-31-28-238 ~]#
```

Check in the browser



## HA-Proxy-Server Steps:

Run following command to Access HA-Proy-Server

- ssh -i your-key.pem ec2-user@Pubipaddress of HA-Proxy-Server instance
- sudo -i
- yum update -y

```
ssh -i mytest_key.pem ec2-user@94.159.48.208
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-172-31-26-216 ~]$ sudo -i
[root@ip-172-31-26-216 ~]# yum update -y
Amazon Linux 2023 kernel livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-26-216 ~]#
```

## - yum install haproxy -y

```
[root@ip-172-31-26-216 ~]# yum install haproxy -y
Last metadata expiration check: 0:00:45 ago on Thu Aug 13 09:33:20 2026.
Dependencies resolved.
=====
Package Architecture Version Repository Size

Installing:
haproxy x86_64 3.0.23-2.amzn2023.0.1 amazonlinux 2.7 M
Transaction Summary

Install 1 Package
Total download size: 2.7 M
Installed size: 8.5 M
Downloading Packages:
haproxy-3.0.23-2.amzn2023.0.1.x86_64.rpm 23 MB/s | 2.7 MB 00:00

Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
 Preparing 1/1
 Running scriptlet: haproxy-3.0.23-2.amzn2023.0.1.x86_64 1/1
 Installing 1/1
 Running scriptlet: haproxy-3.0.23-2.amzn2023.0.1.x86_64 1/1
 Verifying 1/1
 Running scriptlet: haproxy-3.0.23-2.amzn2023.0.1.x86_64 1/1
Installed:
haproxy-3.0.23-2.amzn2023.0.1.x86_64
Complete!
[root@ip-172-31-26-216 ~]# |
```

Vi /etc/hosts

Add sercer1, server2 Public IP's

Vi /etc/ haproxy/haproxy.cfg

Add Server-1, Server-2 public IP's

```
Complete!
[root@ip-172-31-26-216 ~]# vi /etc//hosts
[root@ip-172-31-26-216 ~]# cat /etc//hosts
127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost6 localhost6.localdomain6
54.243.17.85 server-1
98.84.180.46 server-2
[root@ip-172-31-26-216 ~]#
```

- systemctl enable haproxy
- systemctl start haproxy
- systemctl status haproxy

